

Semester Test 3

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Electricity and Electronics EBN 111

27 May 2024

Test Information			
Maximum marks:	46	Full marks:	45
Duration of test:	90 min + (5 min before and 10 min after for OCR)	Open/Closed book:	Closed
Total number of pages (this page included)			18

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Assistant Lecturers: J. Nepembe, S. Twala, L. Hurter

ECSA outcome assessment

The following ECSA graduate attributes are assessed at entry level in this question paper: GA1 and GA2.

Important

1. The examination regulations of the University of Pretoria apply.

2. The departmental rules relevant to electronically graded assessments apply.

3. Answer all questions on the OCR forms provided. The question number 01 in this question paper refers to row 01 of the OCR form, etc.

4. Read the instructions on the OCR form carefully.

5. Answers must include units where applicable.

6. Final answers must be written as real numbers and not as fractions.

7. Use at least 3 significant figures to write numbers, including complex numbers.

8. A complex number may be written in either rectangular or polar format.

INSTRUCTIONS

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form, and also fill in your name and student number on additional pages of the OCR form.

STEP 1 (0 min to 90 min): Do your own calculations on the question paper. Write down the answers for numeric questions on the PRACTICE ANSWER SHEET. Be sure to include the units where applicable.

STEP 2 (90 min to 100 min): Carefully copy the answers to the OCR form.

The Assessment ID is: 2024EBN111S03

Use a mechanical pencil with 0.5 mm HB lead to complete the form.

Each cell should only have one character, and characters may not touch or cross the cell.

Do not use commas! Make use of full stops as decimal points.

Examples for complex numbers on OCR forms

—	5	0	.	9	4	°				
5	∠	—	9	0	°				V	
6	.	9	∠	1	5	.	9	°	μ	A
1	2	—	j	8						
j	1	5							Ω	
—	1	.	5	+	j	8	.	2	m	V

Rectangular:

The j is at the start of the imaginary number.

The j is in the CENTRE of the cell.

The j does NOT cross or touch cell boundaries!

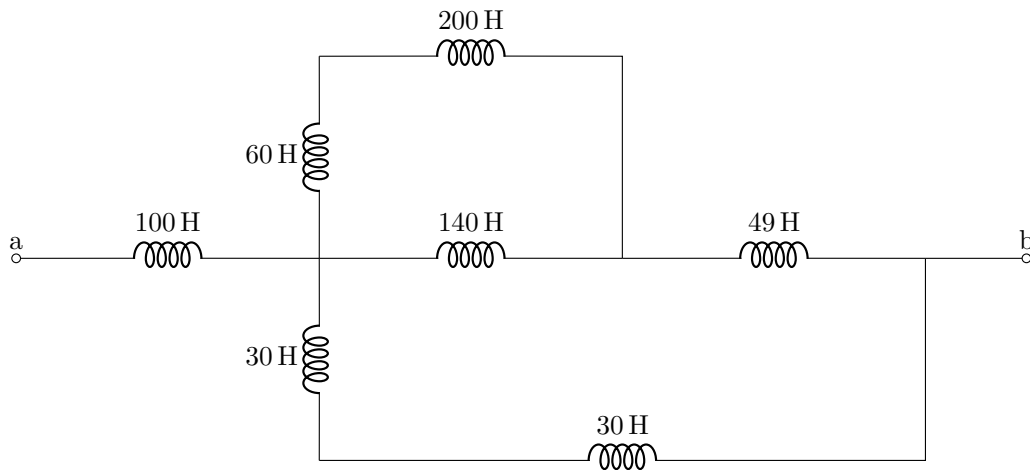
Polar:

The degree sign ° is directly after the angle value.

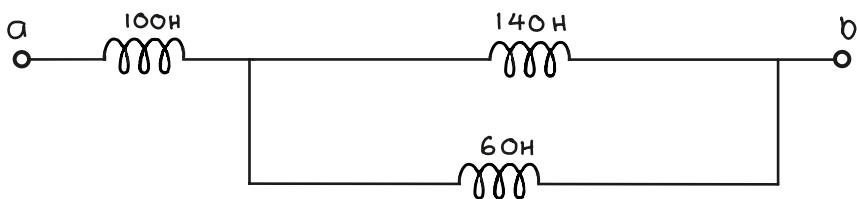
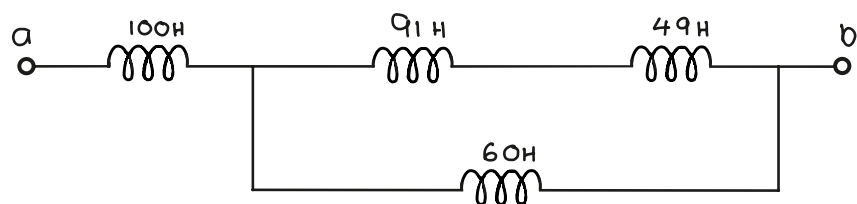
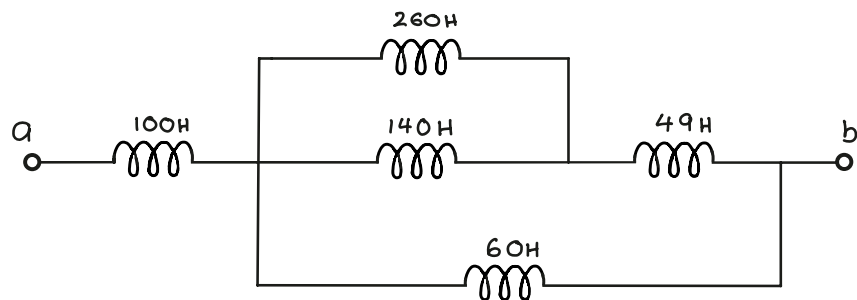
Section 1

Question 1 [2]

Calculate the equivalent inductance L_{ab} with respect to terminals a - b .



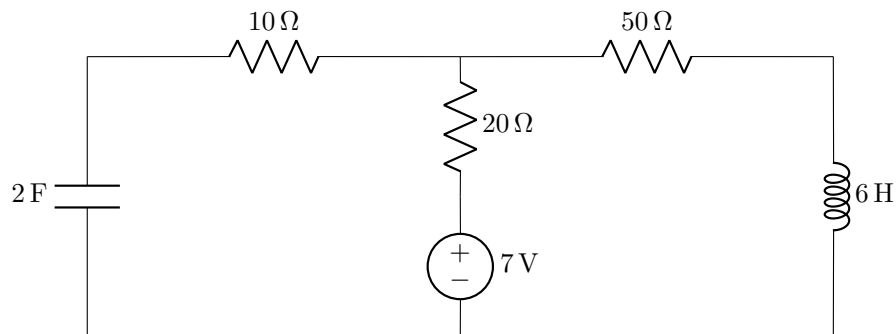
01 L_{ab} [2] 142 H



$$\begin{aligned}
 L_{ab} &= 100\text{ H} + 42\text{ H} \\
 &= \underline{142\text{ H}}
 \end{aligned}$$

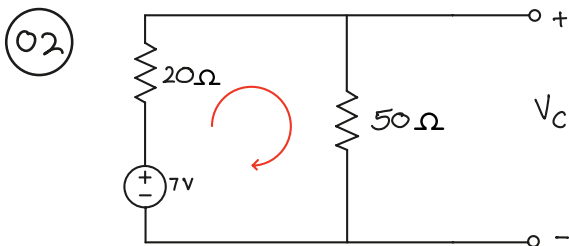
Questions 2 to 3 [4]

The circuit below is under steady state dc conditions. Find the energy w_C stored in the capacitor, and the energy w_L stored in the inductor.



02 w_C [2] **25 J**

03 w_L [2] **30 mJ**



$$i_L = \frac{7}{70} = 0.1 \text{ A}$$

$$V_C = 7 - 20i_L$$

$$= 7 - 20(0.1)$$

$$= 5 \text{ V}$$

$$\therefore w_C = \frac{1}{2} \times 2 \times (5)^2$$

$$= \underline{25 \text{ J}}$$

03

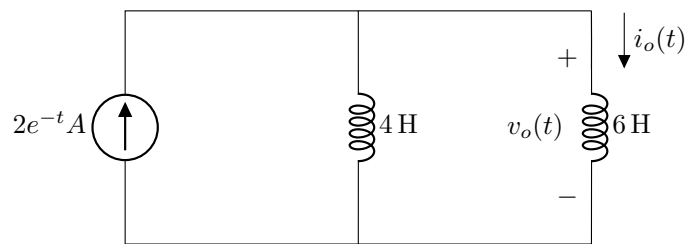
$$w_L = \frac{1}{2} \times 6 \times (0.1)^2$$

$$= 0.03 \text{ J}$$

$$= \underline{30 \text{ mJ}}$$

Questions 4 to 5 [4]

In the circuit below, $i_o(0) = 2$ A. Determine $v_o(t)$ and $i_o(t)$ at $t = 2$ ms.



04 $v_o(2 \text{ ms})$ [2] $-4.79V$

05 $i_o(2 \text{ ms})$ [2] $1.998V$

04 $v_0(t) = Leq \frac{du(t)}{dt}$, where $u(t) = 2e^{-t}$

$$v_0(t) = \left(\frac{6 \times 4}{6 + 4} \right) \frac{d(2e^{-t})}{dt} = 2.4 \times 2 \times \frac{d}{dt}(e^{-t})$$

$$v_0(t) = -4.8e^{-t}V$$

$$v_0(2ms) = -4.8e^{-0.002} = -4.79V$$

05 $i_0(t) = \frac{1}{L_{6H}} \int_0^t v_0(t) + i_0(0)$, where $i_0(0) = 2A$

$$i_0(2ms) = \frac{1}{6} \int_0^t -4.8e^{-t} + 2$$

$$= -\frac{4.8}{6} \int_0^{0.002} e^{-t} + 2$$

$$= -0.8 \left(\frac{e^{-t}}{-1} \Big|_0^{0.002} \right) + 2$$

$$= 0.8(e^{-0.002} - e^{-0}) + 2$$

$$i_0(2ms) = 1.998A$$

Total Section 1: [10]

Section 2

Question 6 [2]

Find $\mathbf{Z}$.

$$\frac{\mathbf{Z} - (6\angle 270^\circ)}{12 + j16} - \frac{\mathbf{Z}}{4} = 1\angle -90^\circ$$

06 $\mathbf{Z}$ [2] $2 + j5\Omega$

$$\textcircled{06} \quad \frac{\mathbf{Z} - (6\angle 270^\circ)}{12 + j16} - \frac{\mathbf{Z}}{4} = 1\angle -90^\circ$$

$$\frac{\mathbf{Z}}{12 + j16} - \frac{(6\angle 270^\circ)}{12 + j16} - \frac{\mathbf{Z}}{4} = 1\angle -90^\circ$$

$$\mathbf{Z} \left(\frac{1}{12 + j16} - \frac{1}{4} \right) = 1\angle -90^\circ + \frac{6\angle 270^\circ}{12 + j16}$$

$$\therefore \mathbf{Z} = \frac{\left[1\angle -90^\circ + \frac{6\angle 270^\circ}{12 + j16} \right]}{\left[\frac{1}{12 + j16} - \frac{1}{4} \right]}$$

$$= \underline{2 + j5\Omega}$$

Questions 7 to 9 [4]	
Given two sinusoidal signals: $v_1(t) = -4 \cos(\omega t - 130^\circ),$ $v_2(t) = 3 \sin(\omega t + 130^\circ).$ The signals $v_1(t)$ and $v_2(t)$ are represented by phasors $\mathbf{V}_1$ and $\mathbf{V}_2$, respectively.	
07	Find the phase angle θ between the two sinusoids. Write your answer in degrees. (See table with examples on Page 1 about how to enter a number with the unit degree.) [1]
08	Choose the correct statement below. 1. $\mathbf{V}_1$ leads $\mathbf{V}_2$; 2. $\mathbf{V}_2$ leads $\mathbf{V}_1$. [1]
09	Calculate the phasor $\mathbf{V}_T = \mathbf{V}_1 + \mathbf{V}_2$. [2]

$$V_1(t) = -4 \cos(\omega t - 130^\circ) \Rightarrow V_1(t) = 4 \cos(\omega t - 130^\circ + 180^\circ)$$

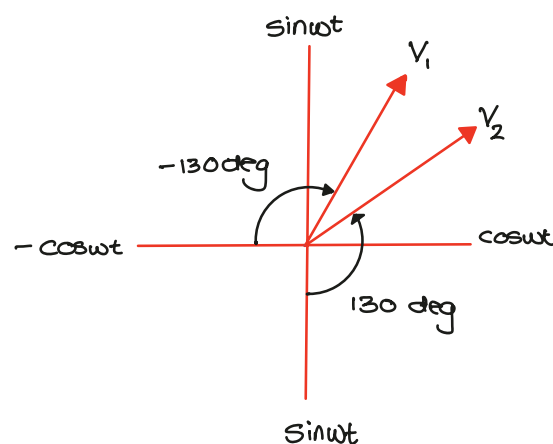
$$\Rightarrow V_1(t) = 4 \cos(\omega t + 50^\circ)$$

$$V_2(t) = 3 \sin(\omega t + 130^\circ) \Rightarrow V_2(t) = 3 \cos(\omega t + 130^\circ - 90^\circ)$$

$$\Rightarrow V_2(t) = 3 \cos(\omega t + 40^\circ)$$

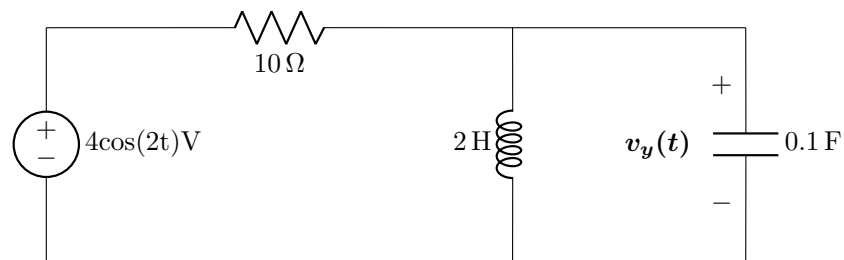
$$\begin{aligned} \textcircled{07} \quad \Delta \theta &= \theta_{V_1} - \theta_{V_2} \\ &= 50^\circ - 40^\circ \\ &= 10^\circ \end{aligned}$$

$$\textcircled{08} \quad V_1 \text{ leads } V_2 \text{ by } 10^\circ$$



$$\begin{aligned} \textcircled{09} \quad V_T &= 4 \angle 50^\circ + 3 \angle 40^\circ \\ &= 6.973 \angle 45.716^\circ \\ &= \underline{6.97 \angle 45.72^\circ \text{ V}} \end{aligned}$$

Question 10 [2]

Find the phasor voltage V_y .10 V_y [2] $3.2 + j1.6 \text{ V}$

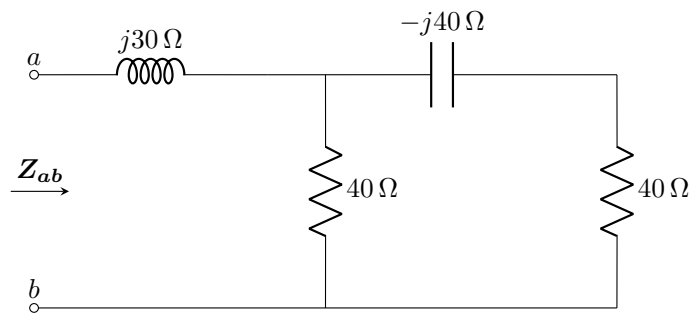
$$\textcircled{10} \quad Z_L = j\omega L = j(2)(2) \\ = j4 \Omega$$

$$Z_C = \frac{1}{j\omega C} = \frac{1}{j(2)(0.1)} \\ = \frac{1}{j(0.2)} \\ = -j5 \Omega$$

$$Z_{\text{par}} = Z_L \parallel Z_C \\ = \frac{(Z_L \times Z_C)}{Z_L + Z_C} \\ = \frac{(j4)(-j5)}{j4 - j5} \\ = j20 \Omega$$

$$V_y = \frac{(4 \angle 0^\circ)(j20)}{(10 + j20)} \\ = \underline{3.2 + j1.6 \text{ V}} \rightarrow$$

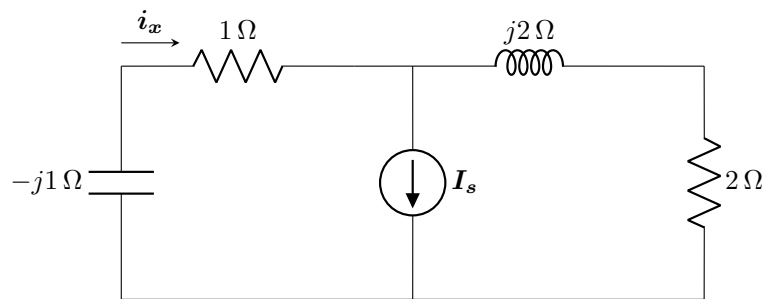
Question 11 [2]



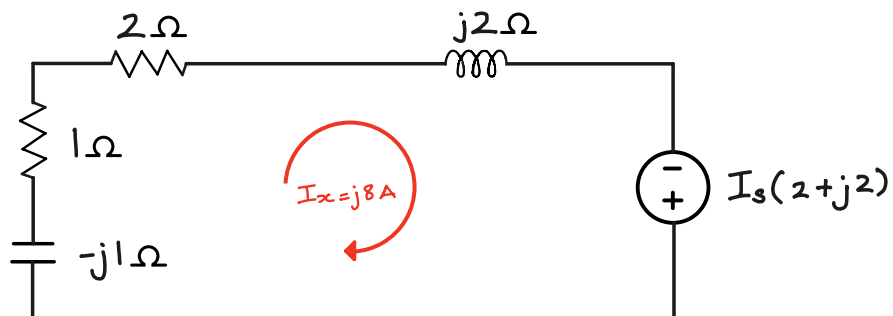
11 Determine Z_{ab} [2] $24 + j22\ \Omega$

$$\begin{aligned} Z_{ab} &= [40 // (40 - j40)] + j30 \\ &= \frac{(40)(40 - j40)}{40 + (40 - j40)} + j30 \\ &= 24 - j8 + j30 \\ &= \underline{24 + j22\ \Omega} \end{aligned}$$

Question 12 [2]



12 Determine I_s if $i_x = 8\angle 90^\circ$ A [2] $4 + j8$ A



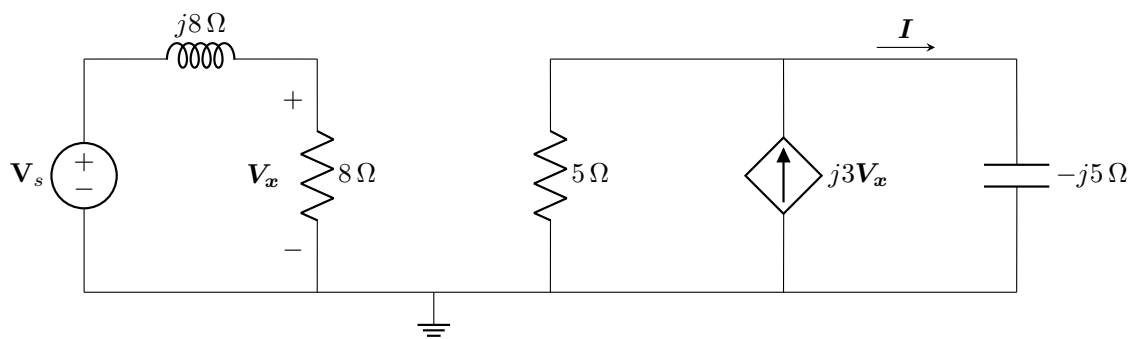
KVL:

$$(1 + 2 - j1 + j2)(j8) - (2 + j2)I_s = 0$$

$$-8 + j24 = (2 + j2)I_s$$

$$I_s = \underline{4 + j8 \text{ A}}$$

Questions 13 to 14 [4]

Given $V_x = -j2$ V, find V_s and I .

13	V_s [2]	$2 - j2$ V
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14	I [2]	$3 + j3$ A
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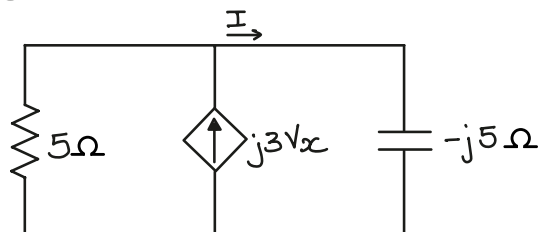
(13) KVL: $-V_s + V_x + (j8)(I_x) = 0$ • $V_x = -j2$ V

$\Rightarrow V_s = V_x + (j8)(I_x)$

$= -j2 + (j8)\left(\frac{-j2}{8}\right)$ $\Rightarrow I_x = \frac{V_x}{8}$

$\Rightarrow V_s = 2 - j2$ V $= \frac{-j2}{8}$

(14)

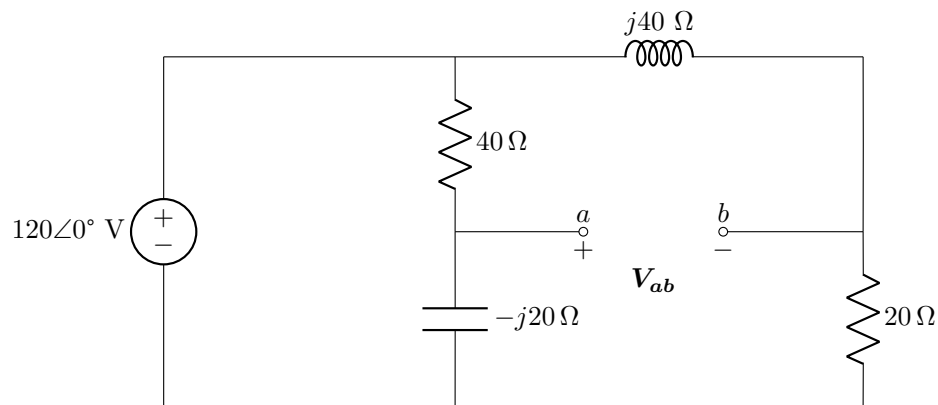
Current Division

$$I = (j3V_x) \left(\frac{5}{(5 - j5)} \right)$$

$$= (j3 \times -j2) \left(\frac{5}{(5 - j5)} \right)$$

$$= 3 + j3 \text{ A}$$

Question 15 [2]

**15** Find V_{ab} [2]

Voltage divisions:

$$V_{ab} = V_a - V_b = 120\angle 0^\circ \cdot \left(\frac{-j20}{40 - j20} - \frac{20}{20 + j40} \right)$$

$$= 0 \text{ V.}$$

Total Section 2: [18]

Section 3

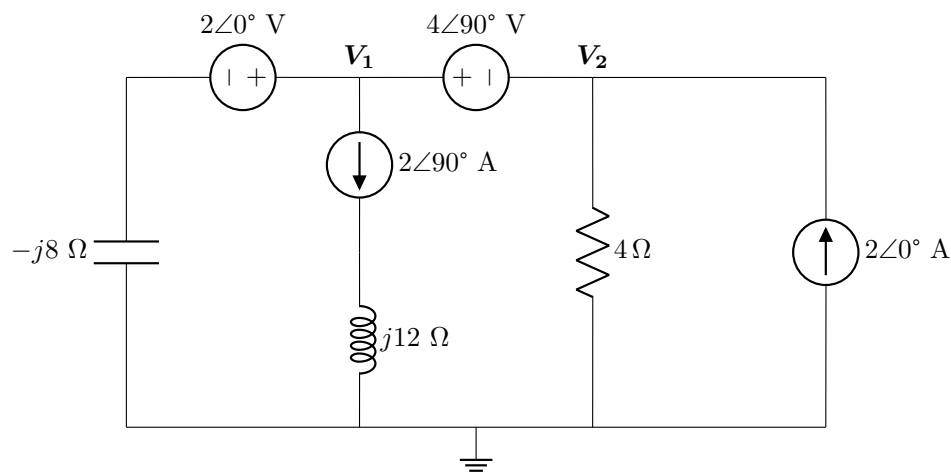
Questions 16 to 19 [4]

Use nodal analysis to setup two equations of the following form:

$$V_1 + aV_2 = b$$

$$cV_1 + -j2V_2 = d$$

Find the unknown parameters of **a**, **b**, **c**, and **d**.



16	a [1]	-1
17	b [1]	j4
18	c [1]	1
19	d [1]	-14 - j16

$$V_1 - V_2 = 4 \angle 90^\circ$$

$$V_1 - V_2 = j4 \text{ V}$$

$$V_1 = j4 + V_2 \quad (1)$$

$$a = -1$$

$$b = j4$$

KCL

$$\left(\frac{V_1}{-j8} \right) - j0.25 + j2 + \frac{V_2}{4} - 2 = 0 \quad (2)$$

$$V_1 - 2 + 16 - j2V_2 + j16 = 0$$

$$V_1 - j2V_2 = -14 - j16$$

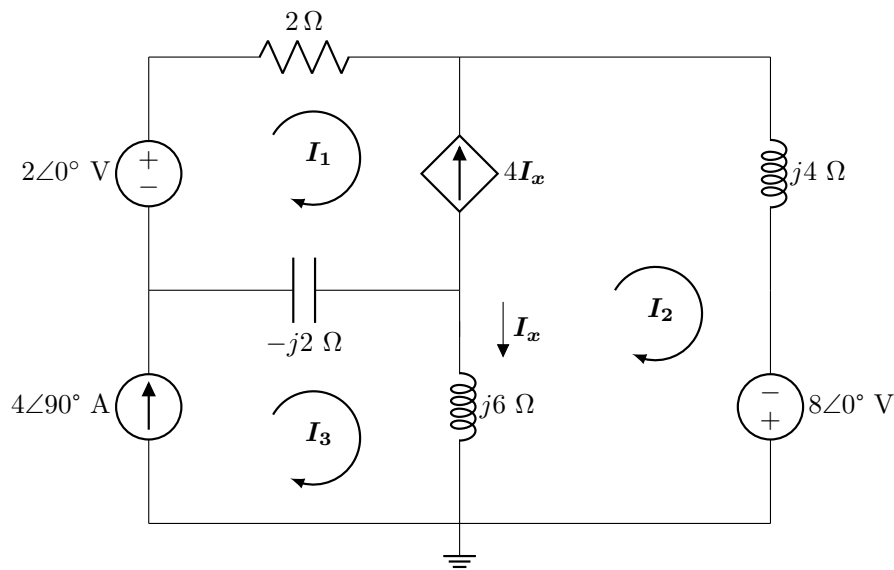
Questions 20 to 23 [4]

Use mesh analysis to set up the following two equations:

$$eI_1 + fI_2 = j16$$

$$gI_1 + j10I_2 = h.$$

Find the unknown parameters e , f , g , and h .

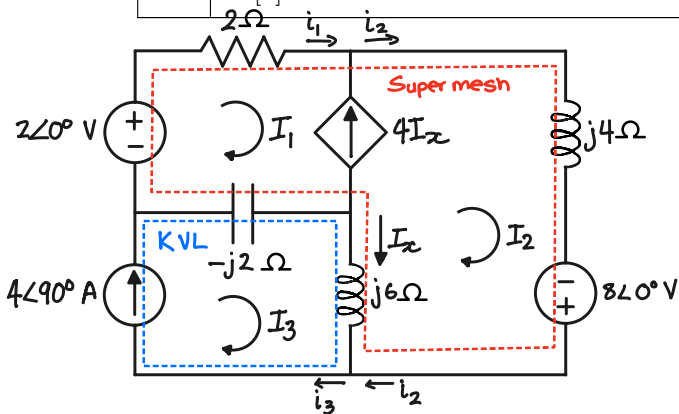


20 e [1] -1

21 f [1] 5

22 g [1] $2 - j2$

23 h [1] -6



$I_3 = 4\angle 90^\circ$ common source

Supermesh

$* i_2 = 4I_x + i_1$

$$0 = -2 + 2I_1 + j4I_2 - 8 + j6(I_2 - I_3) - j2(I_1 - I_3)$$

$$0 = -2 + 2I_1 + j4I_2 - 8 + j6I_2 - j6I_3 - j2I_1 + j2I_3$$

$$-10 + I_1(2 - j2) + j10I_2 - j4I_3 = 0$$

$$\Rightarrow (2 - j2)I_1 + j10I_2 = 10 + j4(4\angle 90^\circ)$$

$$\Rightarrow (2 - j2)I_1 + j10I_2 = -6$$

$$\therefore g = 2 - j2 \quad h = -6$$

$$I_3 = 4\angle 90^\circ \text{ A} = j4 \text{ A}$$

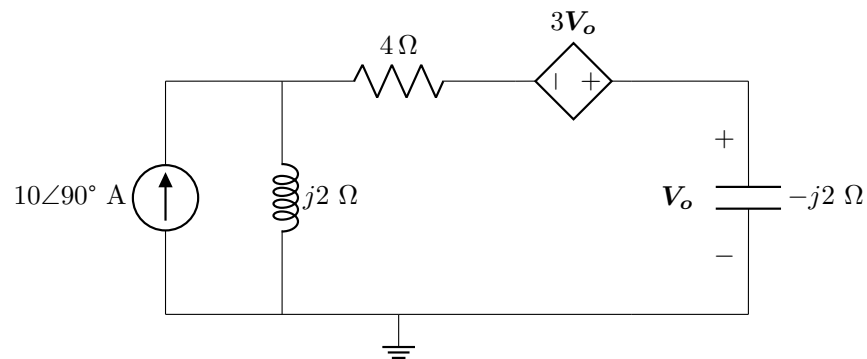
$$\left. \begin{aligned} I_2 - I_1 &= 4I_x \\ I_3 - I_2 &= I_x \end{aligned} \right\} \Rightarrow \begin{aligned} I_2 - I_1 &= 4(j4 - I_2) \\ -I_1 + 5I_2 &= j16 \end{aligned}$$

$$\begin{aligned} e &= -1 \\ f &= 5 \end{aligned}$$

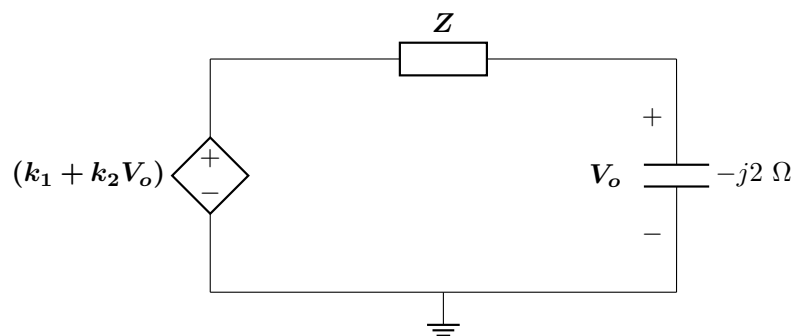
Questions 24 to 25 [2]

Use source transformation to transform Circuit A to Circuit B. Find k_1 and k_2 .

Circuit A:



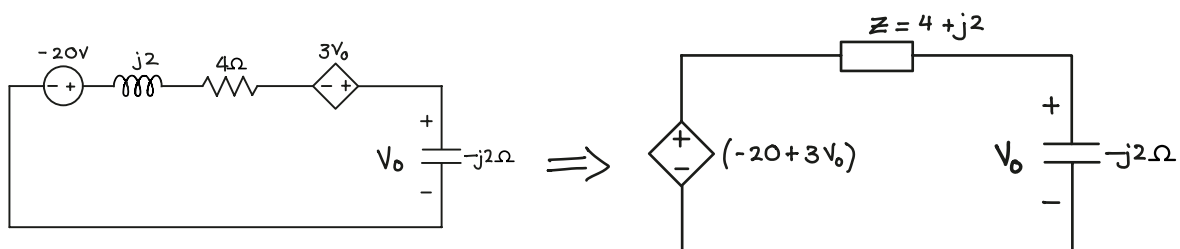
Circuit B:



24	k_1 [1]	-20
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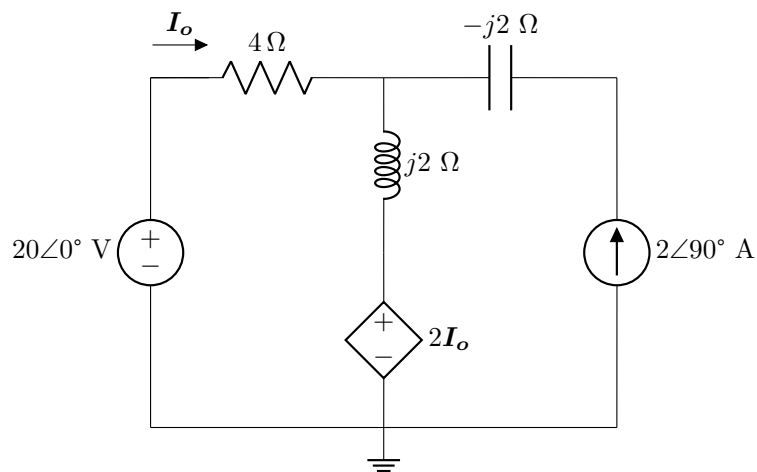
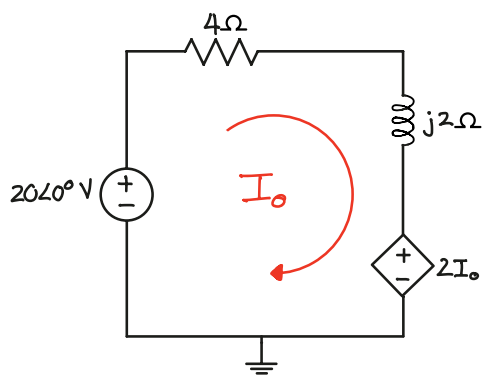
25	k_2 [1]	3
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$$\begin{aligned}
 V &= I \times Z \\
 &= (10 \angle 90^\circ)(j2) \\
 &= -20V
 \end{aligned}$$



$$k_1 = -20$$

$$k_2 = 3$$

Question 26 [2]Find the exclusive contribution I_o^1 from the independent voltage source to I_o .26 | I_o^1 [2] $3 - j$ A

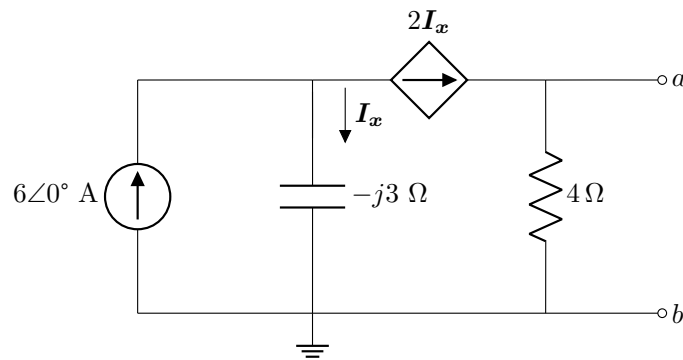
KVL:

$$-20 + 4I_o + j2(I_o) + 2I_o = 0$$

$$I_o = \frac{20}{(6+j2)}$$

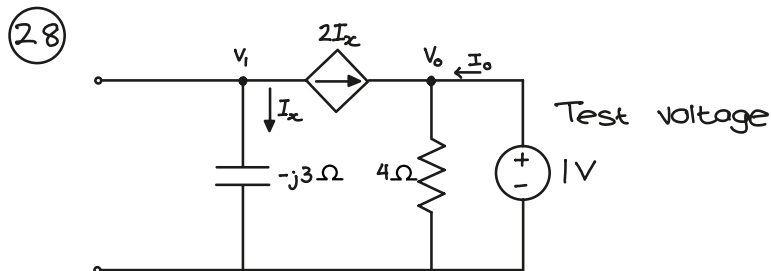
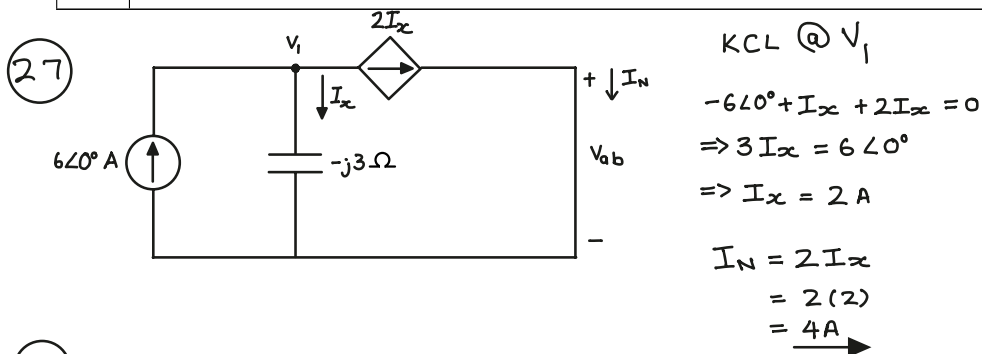
$$= 3 - j \text{ A}$$

Questions 27 to 28 [4]

Find the Norton equivalent current I_N and Norton equivalent impedance Z_N at terminals $a - b$.

27 I_N [2] 4 A

28 Z_N [2] 4 Ω



$$I_x = -2I_x = 0$$

$$\therefore -I_o + \frac{V_o}{4} - 2I_x = 0$$

$$I_o = \frac{V_o}{4} - 2I_x$$

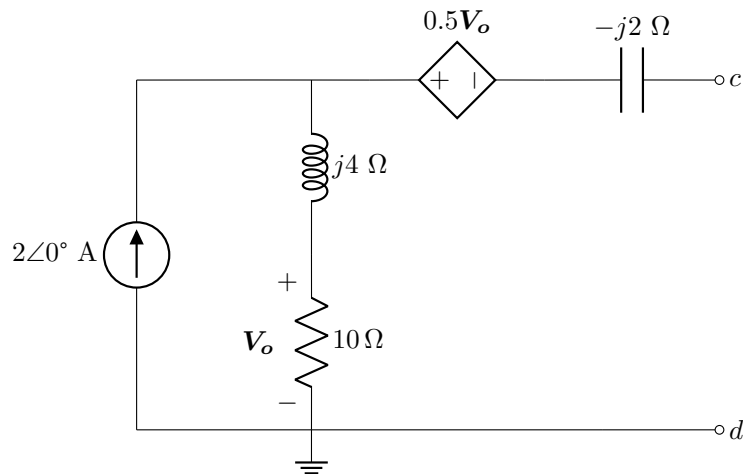
$$I_o = \frac{1}{4} - 0$$

$$I_o = 0.25\text{ A}$$

$$Z_N = \frac{1}{I_o}$$

$$= \frac{1}{0.25}$$

$$Z_N = 4\Omega$$

Question 29 [2]Calculate the Thévenin equivalent voltage V_{Th} at terminals $c - d$.

29	V_{Th} [2]	$10 + j8 \text{ (V)}$
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$$\begin{aligned}
 V_{Th} &= 2\angle 0^\circ \times (10 + j4) - 0.5V_o \\
 &= 2(10 + j4) - 0.5 \times 2 \times 10 \\
 &= 10 + j8 \text{ (V)}
 \end{aligned}$$

Total Section 3: [18]

Grand Total : [46]

PRACTICE ANSWER SHEET

Assessment ID	2	0	2	4	E	B	N	1	1	1	S	0	3
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Round irrational numbers to a minimum of 3 significant figures.

Answer		Unit
Section 1		
01	$L_{ab} = 142$	H
02	$w_C = 25$	J
03	$w_L = 30$	mJ
04	$v_o(2 \text{ ms}) = -4.79$	V
05	$i_o(2 \text{ ms}) = 1.998$	V
Section 2		
06	$Z = 2 + j5$	No unit required
07	$\theta = 10^\circ$	
08	Correct statement is: V_1 leads V_2	No unit required
09	$V_T = 6.97 \angle 45.72^\circ$	No unit required
10	$V_y = 3.2 + j1.6$	V
11	$Z_{ab} = 24 + j22$	Ω
12	$I_s = 4 + j8$	A
13	$V_s = 2 - j2$	V
14	$I = 3 + j3$	A
15	$V_{ab} = 0$	V
Section 3		
16	$a = -1$	No unit required
17	$b = j4$	No unit required
18	$c = 1$	No unit required
19	$d = -14 - j16$	No unit required
20	$e = -1$	No unit required
21	$f = 5$	No unit required
22	$g = 2 - j2$	No unit required
23	$h = -6$	No unit required
24	$k_1 = -20$	No unit required
25	$k_2 = 3$	No unit required
26	$I_o^1 = 3 - j$	A
27	$I_N = 4$	A
28	$Z_N = 4$	Ω
29	$V_{Th} = 10 + j8$	V